

EMCH 260

SOLID MECHANICS

Ming Hu

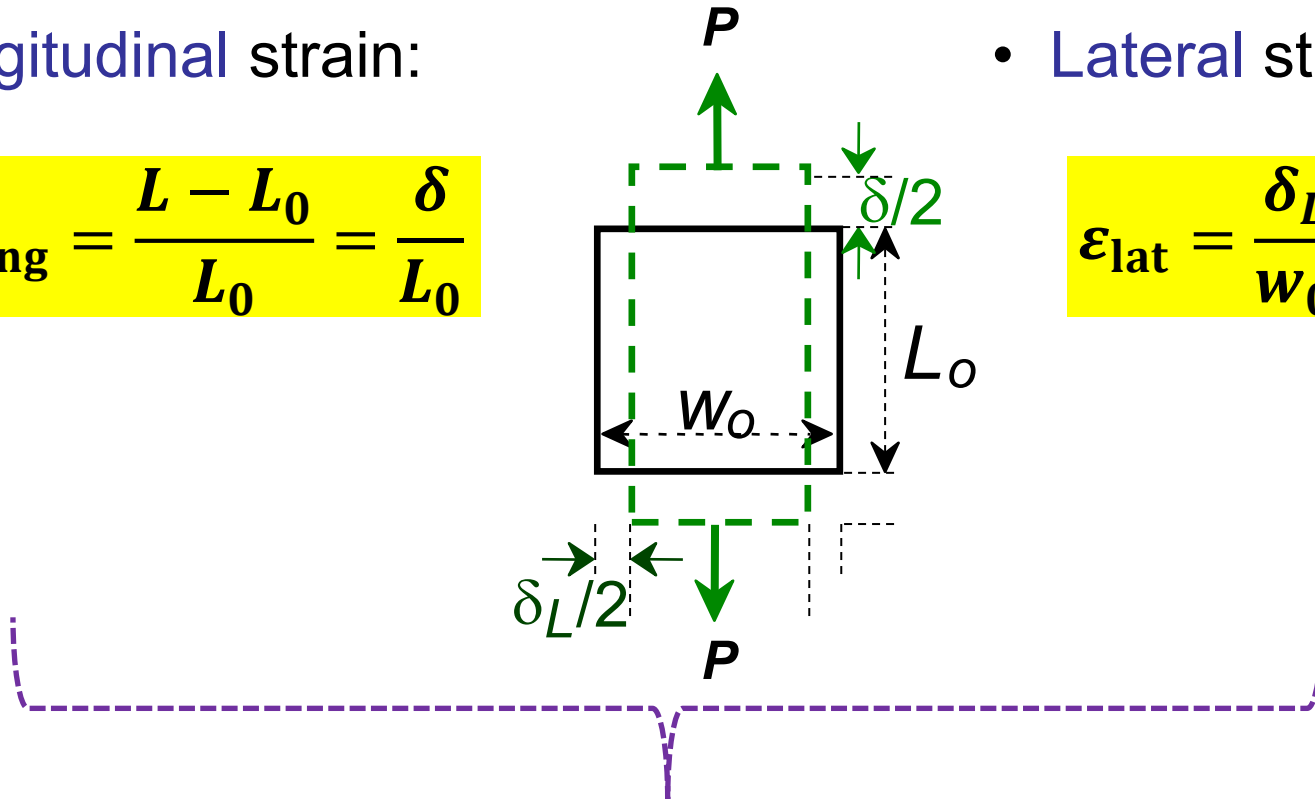
Associate Professor
Department of Mechanical Engineering
University of South Carolina
E-Mail: hu@sc.edu

PREVIOUS ON CHAPTER 2 (STRAIN)

- Loads will cause all material bodies to deform and, as a result, all points in a body will undergo *displacements* or *changes in position*.
- *Normal strain* is a measure per unit length of the elongation or contraction of a small line segment in the body.
- **Longitudinal strain:**
- **Lateral strain:**

$$\epsilon_{\text{long}} = \frac{L - L_0}{L_0} = \frac{\delta}{L_0}$$

$$\epsilon_{\text{lat}} = \frac{\delta_L}{w_0}$$



What's the relationship between them?

Mechanical Properties of Materials

3

Chapter 3.2

- ✓ Understand definition of Poisson's ratio
- ✓ Know how to use the concept of Poisson's ratio to determine & correlate the deformation of materials in different directions

POISSON'S RATIO

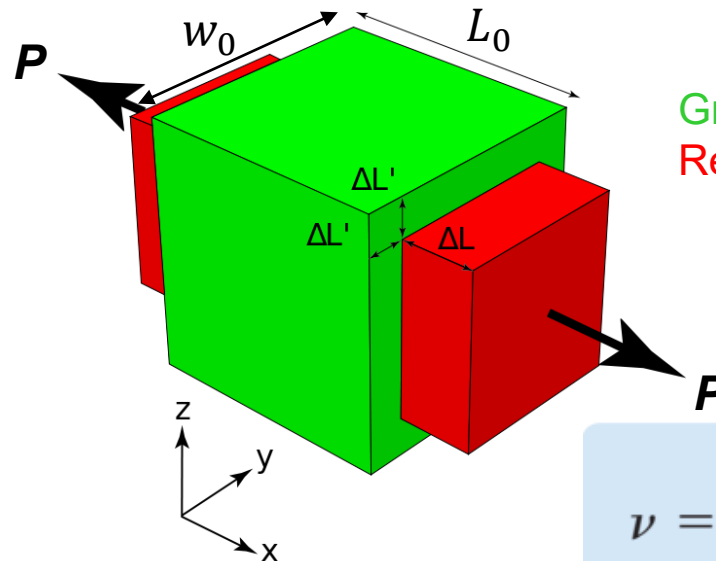
- Uniaxial external force can result in deformation of materials in *both longitudinal and lateral direction simultaneously*.
- However, these strains are usually *not independent*.

Longitudinal strain:

$$\epsilon_{\text{long}} = \frac{\Delta L}{L_0}$$

Lateral strain:

$$\epsilon_{\text{lat}} = \frac{\Delta L'}{w_0}$$



$$\nu = - \frac{\text{lateral strain}}{\text{axial strain}}$$

- **Poisson's Ratio (PR):**

$$\nu = - \frac{\epsilon_{\text{lat}}}{\epsilon_{\text{long}}}$$

Unit: dimensionless

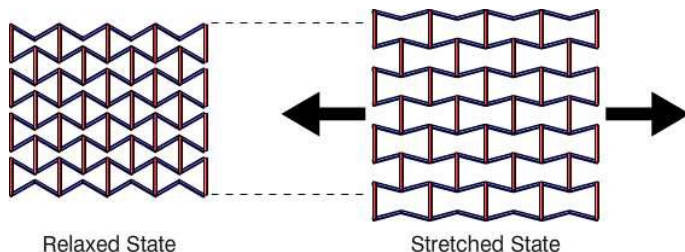
S.D. Poisson (1800s)

SOME NOTES ON POISSON'S RATIO

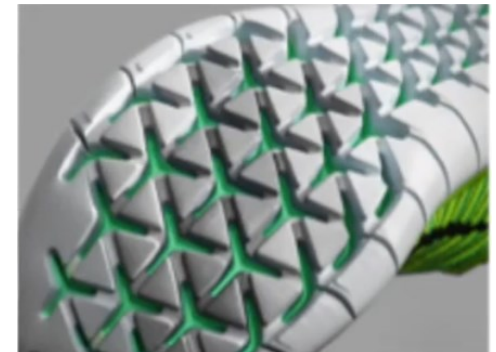
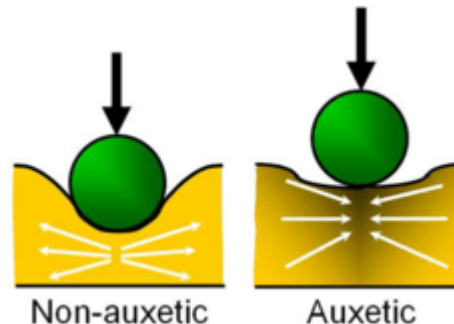
- Within the elastic range, the Poisson's ratio is *constant* and the numerical value is *unique* for a specific material that is *both homogeneous and isotropic*. → intrinsic mechanical property of a material
- The negative sign in the definition indicates that normally *longitudinal elongation* causes *lateral contraction* (negative strain), so that usually you'll get *positive* Poisson's ratio.



Sometimes, the Poisson's ratio could be *negative*! → auxetic materials



<http://www.spaceflight.esa.int/impress/text/education/Glossary/Glossary.html>

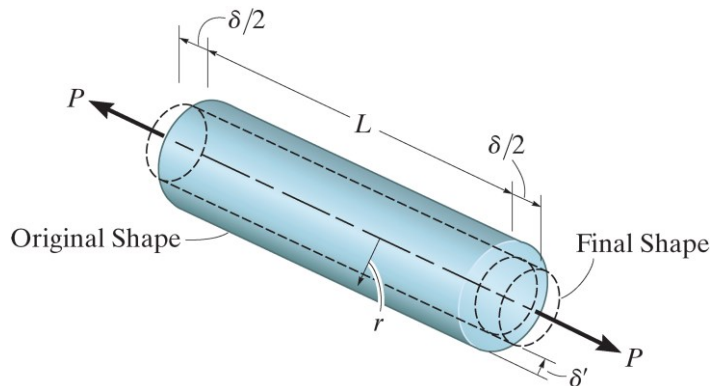


Auxetics have also been found to provide increased fracture toughness and energy absorption.

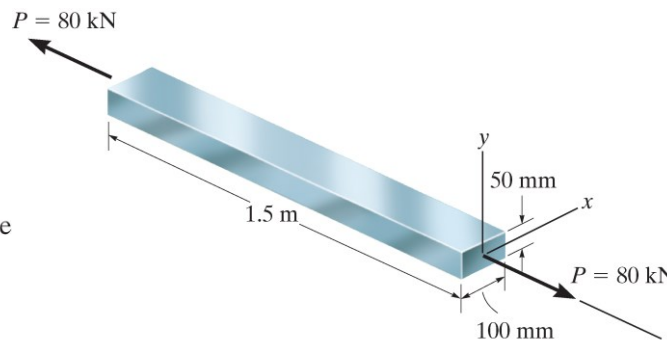
<https://phys.org/news/2015-05-seashell-inspired-material-safety-gear-sport.html>

SOME NOTES ON POISSON'S RATIO

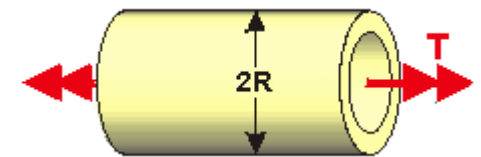
- Generally, $0 \leq \nu \leq 0.5$ (0.5 is the maximum possible value). For most solid materials, the Poisson's ratio can be narrowed down: $0.25 \leq \nu \leq 0.355$.
- With the Poisson's ratio, one can know how the material will deform in the other (perpendicular) direction if the deformation in one direction is known.



Cylinder bar:
longitudinal strain
 $\leftrightarrow$
lateral strain



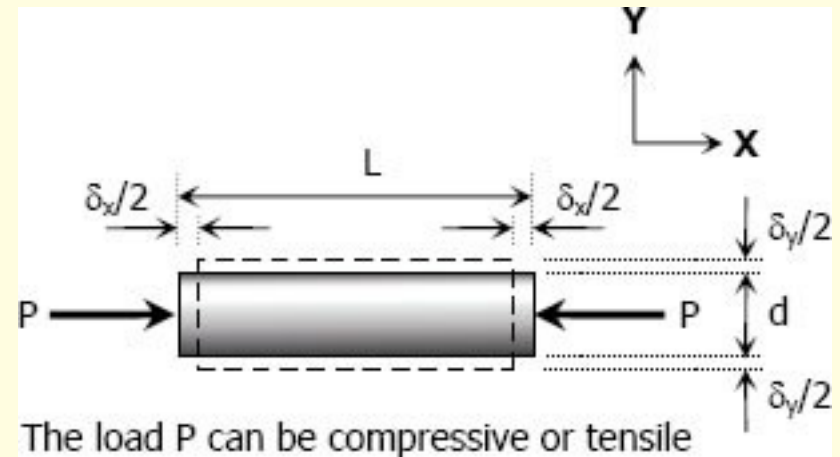
Beam or brick:
longitudinal strain in z-
direction
 $\leftrightarrow$
lateral strains in x- and
y-direction



Hollow tube:
longitudinal strain
 $\leftrightarrow$
lateral strain
(outer and inner
diameter change)

IN-CLASS EXAMPLE 1

A solid cylinder of diameter d carries an axial compressive load P . Assume the Young's modulus of the cylinder is E and the Poisson's ratio is ν . Determine the change in diameter δ .



Poisson's ratio

Logic:

Geometry $\leftrightarrow$ longitudinal and lateral strain $\leftrightarrow$ stress $\leftrightarrow$ external force

Hooke's law $\leftrightarrow$ Modulus of elasticity $\leftrightarrow$ Stress-strain diagram

IN-CLASS EXAMPLE 1 (cont'd)

Solutions

$$\nu = -\frac{\varepsilon_y}{\varepsilon_x}$$

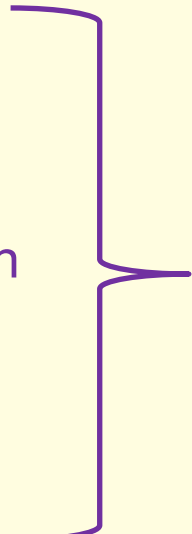
Definition of Poisson's ratio

$$\varepsilon_x = \frac{\sigma}{E} = -\frac{P/(\frac{\pi}{4}d^2)}{E}$$

Hooke's law in axial direction
(define compressive stress as negative)

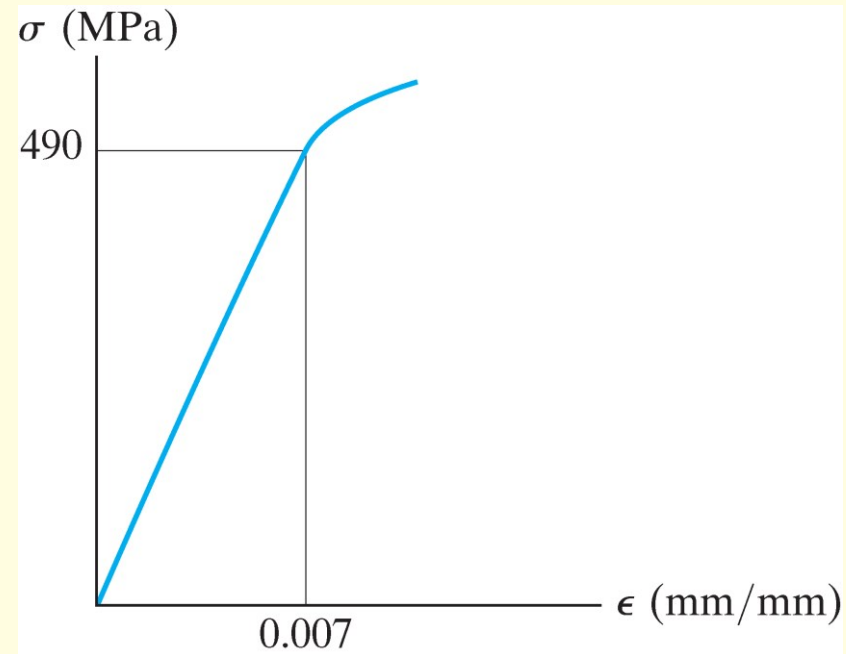
$$\varepsilon_y = \frac{\delta}{d}$$

Definition of lateral strain


$$\delta = \frac{4P\nu}{\pi E d}$$

IN-CLASS EXAMPLE 2

The elastic portion of the stress-strain diagram for an aluminum alloy is shown in the figure. The specimen has an original diameter of 12.7 mm and a gage length of 50.8 mm. If a load of $P = 60$ kN is applied to the specimen, determine its new diameter and length. Take the Poisson's ratio $\nu = 0.35$.



IN-CLASS EXAMPLE 2 (cont'd)

Solutions

- The stress in the specimen:

$$\sigma = \frac{P}{\frac{\pi}{4}d^2} = \frac{60000}{\frac{3.1415926}{4} \times 0.0127^2} = 4.73646 \times 10^8 \text{ Pa}$$

- The Young's modulus of the specimen is determined by the stress-strain diagram:

$$E = \frac{\sigma_0}{\varepsilon_0} = \frac{490 \times 10^6}{0.007} = 7 \times 10^{10} \text{ Pa}$$

- Under external load P , the longitudinal strain is:

$$\varepsilon_{\text{long}} = \frac{\sigma}{E} = \frac{4.73646 \times 10^8}{7 \times 10^{10}} = 0.0067664 \text{ mm/mm}$$

IN-CLASS EXAMPLE 2 (cont'd)

Solutions

- According to the longitudinal strain, the new length of the specimen is:

$$L = (1 + \varepsilon_{\text{long}}) \times L_0 = (1 + 0.0067664) \times 50.8 = 51.1437 \text{ mm}$$

- Then, according to the definition of Poisson's ratio, the lateral strain is:

$$\varepsilon_{\text{lat}} = -\nu \cdot \varepsilon_{\text{long}} = -0.35 \times 0.0067664 = -0.0023682 \text{ mm/mm}$$

(The negative lateral strain means the specimen contracts in the diameter.)

- Finally, according to the definition of the lateral strain, we can determine the new diameter of the specimen:

$$D = (1 + \varepsilon_{\text{lat}}) \times D_0 = (1 - 0.0023682) \times 12.7 = 12.6699 \text{ mm}$$

RELEVANT PRACTICE

- Assignment 1, problem #6 (go to Blackboard under the “Assignments” folder, **Assignment 1 due: 8:30 AM, Monday, Sept. 13th 2021**)
- *Submit your solutions directly onto Blackboard*

OUT-OF-CLASS READING

- **Textbook, pp. 106 – 107**
 - **In particular, Example 3.4 (pp. 107)**